

CUSTOMER CHURN ANALYSIS AND PREDICTION

Prepared By: **Aspita Oluwatoyin**

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1. Introduction

Customer churn is a major challenge for businesses because losing existing customers can significantly impact revenue and growth. This project was conducted to analyze customer behavior, identify factors contributing to churn, and develop a predictive model capable of identifying customers who are likely to leave.

Python was used in Google Colab for data cleaning, analysis, visualization, and machine learning, while Tableau was used to create an interactive dashboard for presenting key insights.

2. Project Objectives

The objectives of this project were to:

- Analyze customer churn behavior.
- Identify key factors influencing customer attrition.
- Perform exploratory data analysis (EDA).
- Build a machine learning model for churn prediction.
- Create a Tableau dashboard to visualize important business insights.

3. Dataset Overview

The dataset contains customer account information, service usage statistics, charges, and customer service interaction records. The target variable, **Churn**, indicates whether a customer stayed with the company or left.

Key features analyzed include:

- International Plan
 - Voice Mail Plan
 - Total Day Minutes
 - Total Day Charges
 - Customer Service Calls
 - Churn Status
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4. Methodology

The project was completed using the following steps:

Data Preparation

- Loaded and inspected the dataset.
- Checked for missing values and duplicates.
- Encoded categorical variables.
- Prepared the data for analysis and modeling.

Exploratory Data Analysis

Several visualizations were created to understand customer behavior, including:

- Customer Churn Distribution
- Correlation Heatmap
- Distribution of Total Day Minutes
- International Plan vs Churn
- Voice Mail Plan vs Churn
- Customer Service Calls vs Churn
- Average Day Charges vs Churn

Machine Learning

A classification model was developed to predict customer churn using customer behavioral and account-related information.

5. Key Findings

The analysis revealed several important insights:

- Customers with international plans showed a higher tendency to churn.
- Customers who frequently contacted customer service were more likely to leave.
- Higher day charges were associated with increased churn rates.
- Customers with voice mail plans generally demonstrated better retention.
- Customer service interactions emerged as one of the strongest indicators of churn risk.

6. Tableau Dashboard

An interactive dashboard was created in Tableau to communicate the findings effectively. The dashboard includes:

- Total Customers KPI
- Total Churned Customers KPI
- Customer Churn Distribution
- International Plan vs Churn
- Voice Mail Plan vs Churn
- Average Day Charges by Churn
- Customer Service Calls vs Churn

The dashboard provides a clear overview of customer behavior and highlights factors associated with churn.

7. Recommendations

Based on the analysis, the following recommendations are suggested:

- Improve customer support quality and responsiveness.
- Monitor customers with frequent customer service interactions.

- Develop targeted retention strategies for customers with international plans.
- Review pricing and service packages for high-charge customers.
- Utilize predictive analytics to identify at-risk customers before they churn.

8. Conclusion

This project successfully analyzed customer churn behavior using data analytics and machine learning techniques. The findings provide valuable insights that can help organizations improve customer retention and make informed business decisions. The Tableau dashboard further enhances decision-making by presenting key insights in a clear and interactive format.

9. Project Links

Google Colab Notebook:

<https://colab.research.google.com/drive/1VjQAI5Djb7j4CPqySI-iboEeZv16HWMZ?usp=sharing>

Tableau Dashboard:

<https://drive.google.com/file/d/1EouPTCRU18JRibiYDdJFGzS0B9uaOy0p/view?usp=sharing>